

Conditional Diffusion Models for Missing ECG Reconstruction

with Predictive Uncertainty Quantification and Global Sensitivity Analysis

Jeevan

September 12, 2026

Abstract

Missing segments in physiological time-series data can arise from sensor disconnection, motion artefacts, transmission failure, poor electrode contact, or temporary acquisition loss. In electrocardiography (ECG), missing intervals are particularly challenging because important waveform morphology may occur entirely inside the unobserved region. This project investigates conditional denoising diffusion probabilistic models (DDPMs) for reconstructing missing ECG segments using the PTB-XL dataset. Conventional linear, cubic-spline and PCHIP interpolation are compared against a basic convolutional autoencoder, a morphology-aware one-dimensional U-Net, and a conditional diffusion model. Models are trained using PTB-XL folds 1–8, validated on fold 9, and tested on a fixed fold-10 benchmark containing missing intervals of 0.25, 0.50, 1.00 and 2.00 seconds. Reconstruction is evaluated using mean absolute error (MAE), root mean squared error (RMSE), waveform correlation and supplementary peak-event recovery metrics.

The diffusion model is additionally used to generate repeated stochastic reconstructions, enabling empirical predictive uncertainty estimation through prediction intervals, prediction interval coverage probability (PICP), mean prediction interval width (MPIW), and the continuous ranked probability score (CRPS). A global uncertainty quantification study is performed using a Polynomial Chaos Expansion (PCE) surrogate and Sobol sensitivity indices to determine the influence of missing-gap duration, gap position, measurement noise and baseline wander on reconstruction quality and predictive uncertainty.

Results show a gap-length-dependent crossover between diffusion and the deterministic U-Net. Diffusion performs best for short gaps, while the U-Net becomes the stronger point estimator for longer gaps. Sensitivity analysis indicates that gap duration and measurement noise dominate predictive uncertainty magnitude, while baseline wander strongly affects probabilistic quality and calibration.

Contents

1	Introduction	4
1.1	Research Questions	4
1.2	Contributions	4
2	Background	5
2.1	ECG Missing-Data Reconstruction	5

2.2	Autoencoders and U-Nets	5
2.3	Diffusion Probabilistic Models	5
2.4	Conditional Diffusion for Imputation	6
3	Dataset and Experimental Design	6
3.1	PTB-XL	6
3.2	Data Split	6
3.3	Normalisation	7
3.4	Missing-Data Simulation	7
3.5	Fixed Fold-10 Benchmark	7
4	Methods	7
4.1	Interpolation Baselines	7
4.2	Basic Convolutional Autoencoder	7
4.3	Morphology-Aware 1D U-Net	8
4.4	Conditional DDPM	8
4.5	Stable Reverse Sampling	8
5	Evaluation Metrics	8
5.1	Mean Absolute Error	8
5.2	Root Mean Squared Error	8
5.3	Pearson Correlation	8
5.4	Peak-Event Recovery	8
6	Reconstruction Results	9
6.1	Basic Autoencoder versus U-Net	9
6.2	Final U-Net versus Diffusion Comparison	9
7	Predictive Uncertainty Quantification	9
8	Global Uncertainty Quantification	10
8.1	Uncertain Inputs	10
8.2	Polynomial Chaos Expansion	10
8.3	Sobol Sensitivity Analysis	11
9	PCE Validation and Sensitivity Results	11
9.1	MPIW90 Sobol Indices	12
9.2	Sample-Uncertainty Sobol Indices	12
10	Paired Statistical Comparison	12
11	Discussion	13
12	Limitations	13
13	Future Work	14

14 Reproducibility	14
15 Recommended Figures	14
16 Conclusion	15

1 Introduction

Physiological time-series signals are frequently incomplete in practical monitoring systems. In ECG acquisition, missing data may occur because of poor electrode contact, motion artefact, wireless transmission failure, sensor detachment, hardware faults, or corruption during storage and preprocessing. Even short gaps can contain important temporal morphology, including QRS complexes and other waveform components. Consequently, simply joining the samples immediately before and after a missing interval may produce a smooth curve without reconstructing the underlying cardiac behaviour.

Traditional interpolation approaches such as linear interpolation, cubic splines and piecewise cubic Hermite interpolation can provide deterministic estimates when gaps are small. However, these methods cannot infer cardiac events that are not already implied by neighbouring samples. Learned reconstruction models offer a possible solution by estimating missing waveform structure from statistical patterns in a training dataset.

This project investigates conditional diffusion models for ECG missing-data reconstruction. Diffusion models are generative latent-variable models that learn to reverse a gradual noising process [1]. Rather than predicting only one reconstruction, a conditional diffusion model can generate multiple plausible trajectories for the same observed context. This is useful for long gaps, where several missing waveforms may be consistent with the observed signal.

1.1 Research Questions

- RQ1:** Can a conditional diffusion model reconstruct missing ECG segments more accurately than interpolation and deterministic neural-network baselines?
- RQ2:** How does reconstruction performance change as missing-gap duration increases from 0.25 to 2.00 seconds?
- RQ3:** Can stochastic diffusion samples provide meaningful predictive uncertainty estimates?
- RQ4:** Which experimental factors most strongly influence reconstruction error, waveform similarity and uncertainty?

1.2 Contributions

- a reproducible fixed-mask Fold-10 ECG benchmark;
- comparison of interpolation, convolutional autoencoder, morphology-aware U-Net and conditional diffusion reconstruction;
- conditional diffusion training with a missing-region noise-prediction loss;
- multi-sample predictive uncertainty analysis;
- PCE surrogate modelling for expensive diffusion experiments;
- Sobol global sensitivity analysis;
- paired statistical comparison of U-Net and diffusion on identical missing intervals.

2 Background

2.1 ECG Missing-Data Reconstruction

An ECG is a structured physiological time series containing repeated but non-identical cardiac events. Missing-data reconstruction is therefore not equivalent to interpolation of a generic smooth curve. In particular, an entire beat may occur inside a missing interval. A useful reconstruction model must infer waveform morphology from temporal context rather than simply connecting the observed endpoints.

2.2 Autoencoders and U-Nets

A deterministic reconstruction model can be written as

$$\hat{x} = f_{\theta}(x_{\text{obs}}, m), \quad (1)$$

where x_{obs} is the observed or corrupted ECG, m is a binary observation mask, and f_{θ} is a neural network. A basic convolutional autoencoder compresses the input into a latent representation and reconstructs the signal. This information bottleneck can encourage overly smooth outputs. A U-Net mitigates this problem through skip connections between matching encoder and decoder resolutions.

2.3 Diffusion Probabilistic Models

Diffusion introduces latent states $x_0, x_1, \dots, x_T$, where x_0 is clean data and x_T approaches Gaussian noise. The forward process is

$$q(x_{1:T} | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}), \quad (2)$$

with

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I). \quad (3)$$

Define

$$\alpha_t = 1 - \beta_t, \quad (4)$$

and

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s. \quad (5)$$

Then a noisy state can be sampled directly:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I), \quad (6)$$

or equivalently

$$\boxed{x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon}, \quad \epsilon \sim \mathcal{N}(0, I). \quad (7)$$

The learned reverse process is

$$p_{\theta}(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \Sigma_{\theta}(x_t, t)). \quad (8)$$

In the DDPM noise-prediction parameterisation, the network predicts

$$\hat{\epsilon} = \epsilon_{\theta}(x_t, t). \quad (9)$$

The implied clean-signal estimate is

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\theta}(x_t, t)}{\sqrt{\bar{\alpha}_t}}. \quad (10)$$

Training uses the simplified objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E} \left[\left\| \epsilon - \epsilon_{\theta} \left(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t \right) \right\|_2^2 \right]. \quad (11)$$

2.4 Conditional Diffusion for Imputation

For ECG imputation, the model is conditioned on observed signal context and a binary mask:

$$\epsilon_{\theta}(x_t, t, x_{\text{observed}}, m). \quad (12)$$

Observed samples are clamped during reverse diffusion:

$$x_t = m \odot x_{\text{observed}} + (1 - m) \odot x_t. \quad (13)$$

The target conditional distribution is therefore

$$p_{\theta}(x_{\text{missing}} \mid x_{\text{observed}}, m). \quad (14)$$

3 Dataset and Experimental Design

3.1 PTB-XL

The project uses the PTB-XL clinical ECG dataset [2]. The current implementation uses Lead II, a sampling frequency of 100 Hz, 10-second recordings and therefore 1000 samples per ECG.

3.2 Data Split

Table 1: PTB-XL split used in the project.

Fold(s)	Purpose
1–8	Training
9	Validation / model selection
10	Final held-out test

3.3 Normalisation

Training-set statistics are estimated from folds 1–8 only:

$$\mu = -0.0014945 \text{ mV}, \quad \sigma = 0.1642238 \text{ mV}. \quad (15)$$

Signals are standardised as

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}. \quad (16)$$

3.4 Missing-Data Simulation

Contiguous missing blocks have durations

$$G \in \{0.25, 0.50, 1.00, 2.00\} \text{ s}, \quad (17)$$

corresponding to 25, 50, 100 and 200 samples at 100 Hz. The observation mask is

$$m_i = \begin{cases} 1, & \text{sample observed,} \\ 0, & \text{sample missing.} \end{cases} \quad (18)$$

3.5 Fixed Fold-10 Benchmark

A frozen manifest stores the exact record, gap length, trial and missing-sample boundaries for each evaluation case. With approximately 2198 Fold-10 records, four gap lengths and five trials per gap length, the benchmark contains approximately

$$2198 \times 4 \times 5 = 43,960 \quad (19)$$

fixed test gaps.

4 Methods

4.1 Interpolation Baselines

Linear interpolation estimates

$$\hat{x}(t) = x_a + \frac{t - t_a}{t_b - t_a} (x_b - x_a). \quad (20)$$

Cubic spline interpolation and PCHIP are also evaluated. PCHIP is included as a shape-preserving alternative to a conventional cubic spline.

4.2 Basic Convolutional Autoencoder

The first learned baseline receives the corrupted ECG and binary mask as two input channels. It learns a deterministic missing-region reconstruction but often oversmooths sharp morphology.

4.3 Morphology-Aware 1D U-Net

The stronger deterministic baseline uses a one-dimensional U-Net with encoder-decoder processing, skip connections and dilated/residual temporal processing. Skip connections preserve high-resolution temporal information and substantially improve morphology recovery over the basic autoencoder.

4.4 Conditional DDPM

The diffusion model receives three channels:

$$[x_t, x_{\text{observed}}, m]. \quad (21)$$

Its noise-prediction loss is restricted to missing samples:

$$\mathcal{L}_{\text{missing}} = \frac{\sum_i (1 - m_i) (\epsilon_i - \epsilon_{\theta,i})^2}{\sum_i (1 - m_i)}. \quad (22)$$

The final training configuration uses 100 diffusion steps, a cosine schedule, learning rate 2×10^{-4} , batch size 64, 30 epochs and EMA decay 0.995.

4.5 Stable Reverse Sampling

Rare numerical explosions in an initial sampler were addressed by explicitly reconstructing $\hat{x}_0$, clipping it to a conservative normalised range, and then evaluating the Gaussian reverse posterior. This stabilised sampling without retraining the network.

5 Evaluation Metrics

5.1 Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|. \quad (23)$$

5.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}. \quad (24)$$

5.3 Pearson Correlation

$$r = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}. \quad (25)$$

5.4 Peak-Event Recovery

A heuristic peak-event metric is used as a supplementary morphology-sensitive measure. It should not be interpreted as a clinically validated R-peak detector.

6 Reconstruction Results

6.1 Basic Autoencoder versus U-Net

For a 1-second gap, the basic autoencoder achieved approximately RMSE 0.1365 and correlation 0.222, whereas the U-Net achieved RMSE 0.0826 and correlation 0.737.

6.2 Final U-Net versus Diffusion Comparison

Table 2: Final Fold-10 comparison. Lower MAE/RMSE is better; higher correlation is better.

Gap	Method	MAE ↓	RMSE ↓	Correlation ↑
0.25 s	U-Net	0.0388	0.0533	0.760
	Diffusion	0.0287	0.0418	0.816
0.50 s	U-Net	0.0440	0.0666	0.794
	Diffusion	0.0372	0.0615	0.815
1.00 s	U-Net	0.0505	0.0826	0.737
	Diffusion	0.0472	0.0846	0.714
2.00 s	U-Net	0.0628	0.1032	0.629
	Diffusion	0.0622	0.1103	0.555

The results show a gap-length-dependent crossover: diffusion is strongest for short gaps, the models are near parity around 1 second, and the deterministic U-Net becomes the stronger point estimator at 2 seconds.

7 Predictive Uncertainty Quantification

Repeated reverse-diffusion samples are generated as

$$x^{(1)}, \dots, x^{(K)} \sim p_{\theta}(x_{\text{missing}} \mid x_{\text{observed}}). \quad (26)$$

The predictive mean is

$$\hat{\mu}_t = \frac{1}{K} \sum_{k=1}^K x_t^{(k)}, \quad (27)$$

and the sample variance is

$$\hat{\sigma}_t^2 = \frac{1}{K-1} \sum_{k=1}^K (x_t^{(k)} - \hat{\mu}_t)^2. \quad (28)$$

A nominal 90% interval uses

$$[L_t, U_t] = [Q_{0.05}, Q_{0.95}]. \quad (29)$$

Prediction interval coverage probability is

$$PICP = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(L_i \leq y_i \leq U_i), \quad (30)$$

and mean prediction interval width is

$$MPIW = \frac{1}{N} \sum_{i=1}^N (U_i - L_i). \quad (31)$$

For a finite ensemble,

$$CRPS = \frac{1}{K} \sum_{k=1}^K |x_k - y| - \frac{1}{2K^2} \sum_{k=1}^K \sum_{\ell=1}^K |x_k - x_\ell|. \quad (32)$$

The dedicated uncertainty experiment uses 500 cases per gap length and 20 diffusion samples per case.

8 Global Uncertainty Quantification

8.1 Uncertain Inputs

The global UQ study varies

$$X = [G, P, N, B], \quad (33)$$

where G is gap duration, P relative gap position, N measurement-noise standard deviation, and B baseline-wander amplitude.

Table 3: Uncertain-factor ranges.

Factor	Symbol	Range
Gap duration	G	$[0.25, 2.00]$ s
Relative gap position	P	$[0, 1]$
Measurement noise	N	$[0, 0.05]$ mV
Baseline wander amplitude	B	$[0, 0.10]$ mV

The final study uses 256 design points, eight Fold-10 ECG records and 10 diffusion samples per design point.

8.2 Polynomial Chaos Expansion

A PCE approximates an expensive scalar output $Y = f(X)$ by

$$\hat{Y} = \sum_{\alpha} c_{\alpha} \Psi_{\alpha}(X). \quad (34)$$

Independent uniform factors are mapped to $[-1, 1]$ and represented using Legendre polynomial basis functions. For degree $p = 2$ and four factors, the basis contains

$$\binom{4+2}{2} = 15 \quad (35)$$

terms.

8.3 Sobol Sensitivity Analysis

The response is decomposed as

$$Y = f_0 + \sum_i f_i(X_i) + \sum_{i < j} f_{ij}(X_i, X_j) + \dots, \quad (36)$$

with total variance

$$\text{Var}(Y) = \sum_i V_i + \sum_{i < j} V_{ij} + \dots. \quad (37)$$

The first-order Sobol index is

$$S_i = \frac{V_i}{\text{Var}(Y)}, \quad (38)$$

and the total-effect index is

$$S_{T_i} = \frac{V_i + \sum_{j \neq i} V_{ij} + \dots}{\text{Var}(Y)}. \quad (39)$$

9 PCE Validation and Sensitivity Results

Table 4: Cross-validated performance of degree-2 PCE surrogates.

Output	CV R^2	CV RMSE
RMSE	0.508	0.0090
CRPS	0.551	0.0053
PICP90	0.729	0.0704
MPIW90	0.888	0.0104
Sample uncertainty	0.879	0.0041
Correlation	0.456	0.0739

9.1 MPIW90 Sobol Indices

Table 5: Sobol indices for MPIW90.

Factor	S_i	S_{T_i}
Gap duration	0.464	0.472
Gap position	0.009	0.011
Measurement noise	0.482	0.488
Baseline wander	0.035	0.039

9.2 Sample-Uncertainty Sobol Indices

Table 6: Sobol indices for mean diffusion sample uncertainty.

Factor	S_i	S_{T_i}
Gap duration	0.482	0.491
Gap position	0.010	0.013
Measurement noise	0.463	0.470
Baseline wander	0.033	0.038

Overall, gap duration and measurement noise dominate predictive spread; gap duration dominates waveform correlation; baseline wander strongly influences probabilistic quality and calibration; and gap position has comparatively little influence.

10 Paired Statistical Comparison

For error metrics such as RMSE, diffusion advantage is defined as

$$A_i = RMSE_{U-Net,i} - RMSE_{Diffusion,i}. \quad (40)$$

Thus $A_i > 0$ indicates a diffusion advantage. For correlation,

$$A_i = r_{Diffusion,i} - r_{U-Net,i}. \quad (41)$$

The analysis uses paired bootstrap confidence intervals, Wilcoxon signed-rank tests, Holm-Bonferroni correction, paired win rates and matched-pairs rank-biserial effect sizes.

Table 7: Fraction of paired cases in which diffusion achieves lower RMSE.

Gap	Diffusion win rate
0.25 s	76.8%
0.50 s	67.1%
1.00 s	50.2%
2.00 s	38.6%

Because the paired sample size is large, effect sizes, confidence intervals and win rates are interpreted alongside p-values rather than significance alone.

11 Discussion

The principal finding is that the benefit of diffusion depends on the amount of missing information. For short gaps, surrounding context strongly constrains the missing ECG and diffusion averaging produces accurate point estimates. For longer gaps, the exact cardiac trajectory is increasingly underdetermined. A stochastic model may generate a plausible but temporally shifted beat, which can receive a worse pointwise RMSE despite being physiologically plausible.

This explains why the deterministic U-Net becomes superior as a point estimator for 2-second gaps while diffusion retains value as a distributional model. In simplified form:

U-Net: What single reconstruction should be predicted? (42)

Diffusion: What reconstructions are plausible given the observed ECG? (43)

The global sensitivity results support this interpretation. Longer gaps increase uncertainty because a greater part of the underlying cardiac trajectory is unobserved. Measurement noise widens the predictive distribution because the conditioning context becomes less reliable. Baseline wander affects probabilistic quality and coverage calibration.

12 Limitations

- Only Lead II is modelled, although PTB-XL contains 12 leads.
- Missingness is synthetic and contiguous rather than taken from real acquisition failures.
- The peak-event metric is heuristic rather than a clinically validated R-peak detector.
- Diffusion inference is computationally more expensive than a single deterministic U-Net pass.
- Repeated diffusion samples represent conditional predictive variability, not complete Bayesian epistemic uncertainty.
- PCE fidelity differs by output, so exact Sobol values for lower- R^2 surrogates should be interpreted cautiously.

- Measurement-noise and baseline-wander ranges are controlled assumptions rather than universal clinical ranges.

13 Future Work

- multi-lead ECG reconstruction;
- evaluation on real dropout and artefact patterns;
- clinically validated morphology metrics;
- downstream rhythm or diagnostic evaluation;
- faster diffusion samplers;
- explicit epistemic uncertainty via ensembles or Bayesian approximations;
- uncertainty calibration;
- comparison with VAEs, masked transformers, state-space models and flow-matching methods.

14 Reproducibility

The project uses fixed train/validation/test folds, training-only normalisation statistics, a permanent Fold-10 missing-gap manifest, fixed random seeds and common random numbers for the global UQ design.

The principal experiment sequence is:

```
python experiments/compute_train_stats.py
python experiments/interpolation_baseline.py
python experiments/train_autoencoder.py
python experiments/evaluate_autoencoder.py
python experiments/train_unet.py
python experiments/create_test_mask_manifest.py
python experiments/evaluate_fixed_manifest.py
python experiments/train_diffusion.py
python experiments/evaluate_diffusion.py --samples-per-input 5
python experiments/evaluate_uncertainty.py
python experiments/run_uq_design.py --design-points 256 --records 8 --diffusion-samples 10
python experiments/analyze_uq_pce.py --degree 2
python experiments/compare_unet_diffusion_stats.py
```

15 Recommended Figures

For the final report, create a folder named `figures` in Overleaf and upload the selected result images. Example:

```

\begin{figure}[H]
\centering
\includegraphics[width=0.9\textwidth]
{figures/diffusion_reconstruction_visualization.png}
\caption{Ground truth, U-Net reconstruction, diffusion mean
and empirical predictive interval.}
\label{fig:diffusion_reconstruction}
\end{figure}

```

Recommended figures include diffusion reconstruction with uncertainty bands, paired RMSE win rate, paired RMSE advantage, paired correlation advantage, and selected Sobol plots for MPIW90 and sample uncertainty.

16 Conclusion

This project evaluated conditional diffusion models for missing ECG reconstruction and compared them against conventional interpolation, a convolutional autoencoder and a morphology-aware one-dimensional U-Net. Conditional diffusion achieved the strongest point reconstruction for short gaps and additionally provided a distribution of plausible missing waveforms from which predictive uncertainty could be estimated. For longer gaps, the deterministic U-Net became more accurate according to RMSE and correlation, demonstrating that probabilistic generative reconstruction and optimal deterministic point prediction are not identical objectives.

Global uncertainty quantification showed that gap duration and measurement noise strongly influence predictive spread, while baseline wander affects calibration and probabilistic reconstruction quality. Overall, the project demonstrates the value of combining generative physiological time-series reconstruction with explicit uncertainty quantification rather than reporting reconstruction accuracy alone.

References

- [1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851, 2020.
- [2] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Frank I. Lunze, Wojciech Samek, and Tobias Schaeffter. Ptb-xl, a large publicly available electrocardiography dataset. *Scientific Data*, 7(1):154, 2020. doi: 10.1038/s41597-020-0495-6.